

Lead Demonstrator: _____ Partner(s): _____ Period: _____ Date: _____

MISSION BRIEFING: Plan, engineer, and defend a live, repeatable demonstration proving **Newton's First Law of Motion** (an object maintains constant velocity unless acted on by a net external force: $\Sigma F = 0$). Debunk common misconceptions using Free-Body Diagrams (FBDs), Claim-Evidence-Reasoning (CER), and automotive engineering safety.

PHASE 1 Select Inertia Category & Codename

☐ **1. Inertia of Rest**
Stationary mass stays at rest when supporting surface accelerates (e.g. tablecloth slip, coin-beaker snap, hoop nut drop).

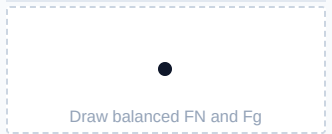
☐ **2. Inertia of Motion**
Moving mass continues straight at constant speed when carrier abruptly stops (e.g. crash cart, skateboard curb halt).

☐ **3. Inertia of Direction**
Circular mass flies along straight tangent line the instant inward tether is cut (e.g. string cut, pie-pan marble exit).

Demonstration Title / Codename: _____

PHASE 5 Free-Body Diagram (FBD) Defense: Proving Net Force

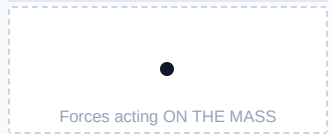
Diagram A: Initial State
 $\Sigma F = 0$ (Equilibrium)



Draw balanced FN and Fg

Vertical: $F_N = F_g$ (cancels). Horizontal: $F_{net} = 0\text{ N} \rightarrow$ constant velocity ($v = 0$ or v_0).

Diagram B: Split-Second Action
 $F_{fwd} = 0\text{ N}$ (No Push!)



Forces acting ON THE MASS

⚠ Zero forward force! Inertia is resistance to acceleration, NOT an active force.

PHASE 2 Bill of Materials & Apparatus Inventory

COMPONENT ITEM	PHYSICS ROLE	SOURCE / READY
1. Inertial Mass:	Resists acceleration (large mass = high inertia)	_____ []
2. Support / Carrier:	Surface or cart accelerated or halted	_____ []
3. Impulse Trigger:	Applies quick external force (yank, flick, stop, cut)	_____ []
4. Safety Catch:	Basin/cushion for soft catch (no glass breakables!)	_____ []

PHASE 6 Scientific Defense: Claim - Evidence - Reasoning (CER)

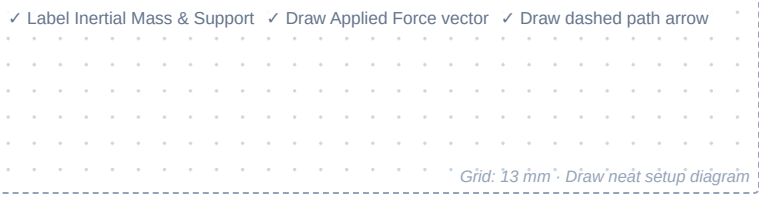
1. CLAIM: _____ State how demonstration proves Newton's First Law.

2. EVIDENCE: _____ Exact physical motion observed during execution.

3. REASONING: Explain using Mass, Inertia, Friction (Static vs. Kinetic), and Net External Force ($F_{net} = 0$).

PHASE 3 Setup Blueprint & Vector Diagram

☒ Label Inertial Mass & Support ☒ Draw Applied Force vector ☒ Draw dashed path arrow



Grid: 13 mm • Draw neat setup diagram

PHASE 7 Automotive Safety & Modern Engineering Connection

System: Automotive 3-Point Inertia-Reel Seatbelts & Pre-Tensioners.

How Inertia Operates:
Passenger stubbornly cruises forward at vehicle speed.

Engineering Solution / Protection:
Inertia-lock mechanism applies backward restraining force.

PHASE 4 Execution Protocol (Step-by-Step)

1. Initial Setup & Alignment

2. The Action (Quick Impulse)

3. Observed Motion (Mass vs Carrier)

4. Troubleshooting & Low Friction

PHASE 8 Demonstration Scoring Checklist

- Apparatus & Protocol (3 pts):** Repeatable setup, safety compliance, clear procedure.
- FBD Vector Accuracy (3 pts):** Proper vectors; zero forward force correctly shown.
- CER Scientific Defense (2 pts):** Valid claim, evidence citations, Newton 1st Law reasoning.
- Real-World Synthesis (2 pts):** Accurate inertia mechanism and life-saving engineering.

TEACHER SCORE
____ / 10